robotic armed vehicle was inducted in January 2019.

SGR-A1, an AI-based sentry gun, is a stationary lethal autonomous weapons system (LAWS). South Korea is using these sentry guns for years along the heavily defended border with North Korea. They are placed at a fixed location to watch and respond to possible threats. These were developed in 2006 and reportedly deployed in 2010 along the demilitarised zone.

A similar system, developed by DoMAAM Super aEgis II, was also unveiled in 2010. It can detect humans at a distance up to 3km during day and 2.2km at night. It is normally equipped with a 12.7mm caliber standard gun.

BrahMos missile (Credit: www.fresherslive.com)

AI in sea

Robots and AI are not only used for research in the deep sea but also for defence applications.

The US defence research agency DARPA has developed an unmanned submarine tracking vessel called Sea Hunter. It has been built with many electronic sensors, image-processing hardware, and software. It can be armed and used for anti-submarine and counter-mine duties. This autonomous ship can travel thousands of miles for months without a human crew.

The Russian Poseidon is an autonomous, nuclear-powered, and nuclear-armed unmanned underwater vehicle under development by Rubin Design Bureau. It is capable of delivering both conventional and nuclear payloads. The Poseidon is a family of drones having multiple features and capabilities. It is an oceanic multipurpose system.

China is believed to have deployed many unmanned underwater vehicles (UUVs) and autonomous underwater vehicles (AUVs) in the South China Sea. The HSU001, one of its autonomous vehicles, was showcased during China's 70th National Day in October last year.

Modernisation of Indian warfare equipment

The government of India has already initiated the process of preparing Indian defence forces in using AI and developing the technology within the country in order to modernise the defence equipment.


HSU001 autonomous underwater vehicles (Credit: www.forbes.com)

A fully indigenous Futuristic Infantry Combat Vehicle (FICV) is being developed by Ordnance Factory Board (OFB) and Defence Research Development Oranisation (DRDO) for use by the Indian Army. Most of the parts to be used in the production will be developed in India. It is expected to be ready within three to five years.

Kamorta-class corvettes with the Indian Navy, currently in service, are a class of anti-submarine warfare corvettes. These were the first anti-submarine warfare stealth corvettes to be built in India. INS Kavaratti corvette is one of the projects that recently completed sea trials and was delivered to the Indian Navy in February 2020.

The Indian Air Force's (IAF) vision for 2020 includes plans to raise its inventory of fighter combat squadrons, acquire more advanced fighters, sophisticated defence systems, and smart long-range weapons. The IAF will receive forty upgraded Su-30MKIs capable of carrying the BrahMos cruise missile, possibly before 2020 end.

BrahMos is a medium-range ramjet supersonic cruise missile that can be launched from submarines, ships, aircraft, or land. It is described as the world's fastest supersonic cruise missile with a high degree of precision and accuracy. It can be used by Indian Army, Navy, and Air Force.

Ghatak, an unmanned combat air vehicle (UCAV) and, Rustom, an unmanned air vehicle (UAV), are under development by DRDO. India is also likely to purchase thirty Reaper drones from the US State Department. EFY